LEAVE SCHEDULER

ABSTRACT

The Leave Scheduler is a web-based application developed to automate and manage the leave process within organizations and educational institutions. The system replaces manual paperwork and email-based approvals with a centralized digital platform. It allows users to apply for leave, view leave balance, track application status, and receive notifications regarding approvals or rejections.

The application supports role-based access control, enabling employees, managers, and administrators to perform specific functions such as leave application, approval management, and report generation. The system automatically updates leave balances according to predefined policies and provides a calendar view of public holidays for better planning.

OBJECTIVES OF THE PROJECT

The main objectives of the Leave Scheduler system are:

1. To automate the leave application and approval process.
2. To provide role-based access control for different users.
3. To maintain accurate leave balance and eligibility records.
4. To implement multi-level approval workflow.
5. To send automated email notifications for each leave action.
6. To generate detailed reports for administrative analysis.
7. To reduce paperwork and administrative workload.
8. To provide a user-friendly interface for easy access and operation.

SYSTEM ARCHITECTURE

The Leave Scheduler follows a three-tier architecture:

1. Presentation Layer (Frontend)

- Developed using ReactJS
- Provides interactive user interface
- Handles user input, form validations, and dashboard display

2. Application Layer (Backend)

- Developed using Spring MVC
- Handles business logic
- Manages authentication, authorization, and workflow processing
- Processes leave calculations and approval logic

3. Data Layer (Database)

- Implemented using MySQL
- Stores user data, leave records, approval status, and reports
- Ensures data integrity and security

This layered architecture ensures scalability, maintainability, and separation of concerns.

MODULES OF THE LEAVE SCHEDULER SYSTEM

1️⃣ User Management Module

- User registration and login
- Role-based access control (Employee, Manager, Admin)
- Profile management

2️⃣ Leave Application Module

- Apply for leave (type, dates, reason)
- View leave history
- Check leave balance
- Withdraw pending leave requests

3️⃣ Leave Approval Module

- Manager reviews leave requests
- Approve or reject applications
- Add remarks
- Update leave status automatically

4️⃣ Reports & Administration Module

- Generate leave reports
- Manage leave policies
- Add holidays
- Monitor system records.

SUMMARY OF TECHNOLOGY STACK

Layer:	Technology Used
Frontend:	ReactJS, HTML, CSS, JavaScript
Backend:	Spring MVC, Java, REST APIs
Database:	MySQL
Tools:	Tomcat, Maven, Git, Postman

CONCLUSION

The Leave Scheduler system successfully automates and streamlines leave management processes in organizations and educational institutions. By implementing role-based access control, automated workflow processing, and real-time leave balance updates, the system improves efficiency, transparency, and accuracy. The use of modern web technologies ensures scalability, security, and maintainability, making it a reliable solution for institutional leave management.